

# JINFAN XIANG

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## EDUCATION

### Columbia University GPA: 3.8/4.0

Aug 2024 - Dec 2025

M.S. Electrical Engineering

New York, NY

Relevant Coursework: Generative AI, Neural Networks & Deep Learning, NLP, Computer Systems for Data Science

### Sichuan University GPA: 3.6/4.0

Aug 2020 - Jul 2024

B.Eng. Measurement Control Technology and Instruments

Chengdu, CN

Relevant Coursework: Data Structure & Algorithm, Data Analysis

## TECHNICAL SKILLS

- Programming Languages: Python, Java, SQL, HTML&CSS, MATLAB
- Databases and Tools: PostgreSQL, Git, Docker
- Frameworks and Platforms: Spring Boot, Django, Flask, RESTful API, React, Node.js, Next.js, OpenFeign, Caffeine, LangChain, Spring Security, AWS, Google Cloud Platform

## WORK EXPERIENCE

### Founder & Software Engineer

Mar 2026 – Present

AtomByte LLC, Remote

- Developed a **multimodal AI journaling platform** leveraging **Next.js**; accelerated full-stack architecture design using AI coding agents and orchestrated **GPT-4o & DALL-E 3** pipelines for generative outputs.
- Engineered a **client-side data visualization** engine to map user mood histories into 2D sentiment clusters, applying **t-SNE dimensionality reduction** to 1536-D OpenAI embeddings while eliminating backend compute costs.
- Architected an **anonymous social system** utilizing a UUID-based "shadow identity" pattern, optimizing PostgreSQL schemas to enable cross-user interactions without traditional authentication flows.

### Backend Development Intern

Jun 2025 - Aug 2025

Springer Capital Investments LLC, Remote

- Led a 3-person intern team to build **Flask-based microservices** for **HubSpot CRM data extraction**, integrating **PostgreSQL** to persist normalized customer and business data.
- Built a **resilient extraction pipeline** with **ThreadPoolExecutor** concurrency, cursor-based pagination handling, and automatic retries, increasing CRM data extraction throughput by **3x**
- Defined RESTful extraction, status, and monitoring APIs with Flask-RESTX and Swagger, aligning 10+ HubSpot CRM properties with internal data requirements to **improve team-level data accessibility**.
- Containerized microservices with **Docker** across dev/test/prod environments, adding health checks, structured logging, and monitoring endpoints to support repeatable releases and reduce environment drift by **90%**.

## PROJECTS

### A Personalized Twitch Resources Recommendation Engine

Feb 2025 – May 2025

- Built a full-stack platform with **Spring Boot and React**, integrating Twitch games and clips APIs through **OpenFeign**.
- Designed **Spring Data JDBC schemas** for users, streaming items, and favorites, enforcing constraints to **prevent duplicate records**.
- Implemented a **favorites-based recommendation service** that converts saved items into game-ID signals and returns grouped stream/video/clip recommendations.
- Secured user login and favorite workflows with **Spring Security session authentication**; added **Caffeine caching** with eviction on favorite updates and documented dockerized AWS deployment.

### NEXT AI: Full-Stack Web-Based Q&A Agent

Sep 2024 - Nov 2024

- Optimized the interface with React, reducing render time by **60%** via **virtualized rendering** and **debounced state updates**.
- Built **Node.js** REST APIs for **chunked PDF uploads**, document-processing status tracking, and Redis caching of frequently accessed query results.
- Developed a **RAG-based Q&A workflow** for document loading, splitting, storage, retrieval, and interactive Q&A responses, integrating **OpenAI GPT-3.5 Turbo** and **LangChain**.

### An Online Accommodation Rental Application

Jul 2022 - Oct 2022

- Developed RESTful APIs for a booking platform, supporting authentication, listings, location search, and reservations.
- Secured guest and host workflows using **JWT-based Spring Security** authentication and **role-based access control**.
- Modeled users, listings, and bookings in PostgreSQL, storing reservation data and listing coordinates for **distance-based property search**.
- Integrated **Google Cloud Storage** for listing images and **Google Maps Geocoding API** for address-to-coordinate conversion.